

Frédéric Andrian

PhD Applicant · AI Safety and Agentic Systems

Bordeaux, France

📞 (+33) 769875327 | ✉ fredericj.andrian@gmail.com | 🌐 AndriantsehoFrederic | 📄 FredericAndriantseho

Research Profile

Research-oriented AI Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. My research internships in agentic AI and reinforcement learning confirmed my desire to remain in research, where I can continually discover new questions, methods, and technical skills. I am particularly motivated by the rapid evolution of AI and by the opportunity to connect ideas from different fields to address complex problems. At INRIA, I independently developed a multi-step LLM agent without an orchestration framework, investigated memory and context management, implemented RAG, and evaluated open-source models and agent platforms. I now seek a PhD focused on multi-agent AI safety, trust, security, and privacy-preserving retrieval. Available for Spring or Fall 2027 entry.

Research Experience

INRIA, Mnemosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step AI agent that helps researchers discover scientific papers.
- Built the agent without an orchestration framework to understand and control its tools, state, memory, context management, and multi-step execution mechanisms.
- Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and the ChromaDB vector database.
- Benchmarked Qwen, Gemma, GLM, and multiple quantization configurations on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Evaluated OpenClaw as an autonomous agent platform within an isolated Docker environment.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js interface that enabled researchers to interact with the agent and its tools.
- Managed a Dockerized web application for using LLMs installed locally on a Mac Studio, including model installation and configuration.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Dockerized selected tools and implemented a GitHub CI/CD pipeline to automate integration.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed, implemented, and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Research and Technical Skills

Agentic AI	Framework-free multi-step agents, tool design, orchestration, memory and context management, OpenAI API, OpenClaw
RAG and Data Science	Chunking, embeddings, semantic search, ChromaDB, MongoDB, SQL, data collection and preprocessing
Model Evaluation	Experimental design, Qwen, Gemma, GLM, quantization, latency, TTFT, response quality, context size
Machine Learning	Python, PyTorch, Deep Learning, Reinforcement Learning, Random Forest, NLP
Software Engineering	FastAPI, REST APIs, OpenAPI, React, Next.js, Linux, Docker, Git/GitHub, CI/CD
Scientific Computing	Slurm, HPC environments, Raspberry Pi deployment
Languages	French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.